

Larissa García Valdivia

Crear un contenedor

```
PS C:\Users\laris> docker run --name mysql2 -e MYSQL_ROOT_PASSWORD=root -p 3306:3306 -d mysql:8
a89c37579f9f9b3f7e8abe2703daee3297204b4fbf81dad7ebc9331c2ff1d968
```

```
Run 'docker run --help' for more information
PS C:\Users\laris> docker run --name mysql2 -e MYSQL_ROOT_PASSWORD=root -p 3306:3306 -d mysql:8
a89c37579f9f9b3f7e8abe2703daee3297204b4fbf81dad7ebc9331c2ff1d968
```

What's next:

Debug this container error with Gordon → docker ai "help me fix this container error"
docker: Error response from daemon: failed to set up container networking: driver failed programming external connectivity on endpoint mysql2 (c416e035fa74d0b9f9e51a64fc4171aa08b371b06b4029dc37d9abd3554602f9): Bind for 0.0.0.0:3306 failed: port is already allocated

Run 'docker run --help' for more information

```
PS C:\Users\laris> docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
a89c37579f9f	mysql:8	mysql2	17 seconds ago	Created	
7ea50c9ae064	mysql:8	mysql	3 minutes ago	Up 3 minutes	0.0.0.0:3306->3306/tcp
33936f703508	mysql	mysql	11 minutes ago	Created	
f9137d853d44	mysql:8	mysql	3 days ago	Exited (255) 52 minutes ago	0.0.0.0:3306->3306/tcp

Prender el contenedor

```
PS C:\Users\laris> docker start mysql2
mysql2
PS C:\Users\laris> docker exec -it a89c37579f9f bash
```

```
bash-5.1# mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 8.4.11 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

Mostrar las bases de datos

```
mysql> show databases;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
4 rows in set (0.03 sec)
```

Crear la base de datos y las tablas

```
mysql> CREATE DATABASE biblioteca;
Query OK, 1 row affected (0.03 sec)

mysql> use biblioteca;
Database changed
mysql> CREATE TABLE IF NOT EXISTS Usuarios (
    ->     id_usuario INT PRIMARY KEY,
    ->     nombre_usuario VARCHAR(100) NOT NULL,
    ->     email_usuario VARCHAR(100) NOT NULL
    -> );
Query OK, 0 rows affected (0.03 sec)
```

```
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE IF NOT EXISTS Libros (
    ->     id_libro INT PRIMARY KEY,
    ->     titulo_libro VARCHAR(150) NOT NULL,
    ->     autor_libro VARCHAR(100) NOT NULL
    -> );
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> CREATE TABLE IF NOT EXISTS Prestamos (
->     id_prestamo INT PRIMARY KEY,
->     fecha_prestamo TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
->     id_usuario INT NULL,
->     id_libro INT NULL,
->     CONSTRAINT fk_prestamos_usuario FOREIGN KEY (id_usuario)
->         REFERENCES Usuarios(id_usuario)
->         ON DELETE SET NULL
->         ON UPDATE CASCADE,
->     CONSTRAINT fk_prestamos_libro FOREIGN KEY (id_libro)
->         REFERENCES Libros(id_libro)
->         ON DELETE SET NULL
->         ON UPDATE CASCADE
-> );
Query OK, 0 rows affected (0.06 sec)
```

Insertar los datos

```
mysql> INSERT INTO Usuarios (id_usuario, nombre_usuario, email_usuario) VALUES
-> (1, 'Larissa', 'larissa@gmail.com'),
-> (2, 'Mafer', 'mafer@gmail.com'),
-> (3, 'Rodin', 'rodin@gmail.com');
Query OK, 3 rows affected (0.04 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> INSERT INTO Libros (id_libro, titulo_libro, autor_libro) VALUES
-> (0, 'Katabasis', 'F. Kuang'),
-> (1, 'Don Quijote de la Mancha', 'Miguel de Cervantes'),
-> (2, 'Cien aos de soledad', 'Gabriel Garca Mrquez');
Query OK, 3 rows affected (0.03 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> INSERT INTO Prestamos (id_prestamo, fecha_prestamo) VALUES
-> (0, '2026-01-01 00:00:00'),
-> (1, '2026-02-28 00:00:00'),
-> (2, '2026-03-14 00:00:00');
Query OK, 3 rows affected (0.03 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

Mostrar todas las tablas

```
mysql> SELECT * FROM Usuarios
```

```
-> ;
```

id_usuario	nombre_usuario	email_usuario
1	Larissa	larissa@gmail.com
2	Mafer	mafer@gmail.com
3	Rodin	rodin@gmail.com

```
3 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Libros
```

```
-> ;
```

id_libro	titulo_libro	autor_libro
0	Katabasis	F. Kuang
1	Don Quijote de la Mancha	Miguel de Cervantes
2	Cien aos de soledad	Gabriel Garca Mrquez

```
3 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Prestamos;
```

id_prestamo	fecha_prestamo	id_usuario	id_libro
0	2026-01-01 00:00:00	NULL	NULL
1	2026-02-28 00:00:00	NULL	NULL
2	2026-03-14 00:00:00	NULL	NULL

```
3 rows in set (0.00 sec)
```